

$$1) m_e = 0$$

$$\sigma_e^2 = M_e = \frac{9}{4} + \frac{9}{4} + \frac{1}{4} + \frac{1}{4} = 5 \quad R_e(f) = 5T$$

$$\bar{R}_v(f) = R_e(f) \cdot \frac{1}{T^2} |G(f)|^2$$

$$\begin{aligned} \bar{M}_v &= \frac{5T}{T^2} \cdot E_g = \frac{5T}{T^2} \cdot \int_{-\infty}^{+\infty} V_0^2 T^2 \text{rect}(fT) df \\ &= \frac{5T^2}{T^2} \cdot V_0^2 = 5V_0^2 \end{aligned}$$

$$2) h(kT) = (0.9)^k 1_0(kT) + (0.1)^k 1_0(kT)$$

$$R_x(f) = \sigma_x^2 \cdot T$$

$$\begin{aligned} R_y(f) &= R_x(f) |H(f)|^2 \\ &= \sigma_x^2 \cdot T \cdot |H(f)|^2 \end{aligned}$$

$$\begin{aligned} M_y &= \int_{-\infty}^{+\infty} R_y(f) df = \int_{-\infty}^{+\infty} \sigma_x^2 T |H(f)|^2 df = \sigma_x^2 \cdot T \cdot E_h \\ &= \sigma_x^2 \cdot T^2 \cdot \sum_{k=-\infty}^{+\infty} |h(kT)|^2 = \sigma_x^2 \cdot T^2 \cdot \sum_{k=0}^{+\infty} (0.9^k + 0.1^k)^2 \end{aligned}$$

$$= \sigma_x^2 T \sum_{k=0}^{+\infty} 0.81^k + 0.01^k + 2 \cdot 0.09^k$$

$$= \sigma_x^2 T \left[\frac{1}{1-0.81} + \frac{1}{1-0.01} + \frac{2}{1-0.09} \right]$$

B) $p(t)$ ha estensione spettrale $(-\frac{2}{T}, \frac{2}{T})$ (conv. di
 $\frac{1}{4} \text{rect}(\cdot)$ con estensione $(-\frac{1}{2T}, \frac{1}{2T})$) uguale a
quella di $h(f) = \frac{h_0 T}{4} \text{rect}(\frac{f}{4/T})$

Dunque

$$C(f) = \frac{h_0 T}{4} G(f)$$

$$c(t) = \frac{h_0 T}{4} \cdot V_0 \text{sinc}^2\left(\frac{t}{T}\right) \quad \text{è di Nyquist, con}$$

$$V_0' = \frac{h_0 T}{4} V_0$$

$$\sigma_m^2 = R_0 \int_{-\infty}^{+\infty} \frac{h_0^2 T^2}{16} \text{rect}\left(\frac{fT}{4}\right) df$$

$$= R_0 \frac{h_0^2 T^2}{16} \cdot \frac{4}{T} = R_0 h_0^2 \frac{T}{4}$$

$$P_e = \frac{4}{3} \cdot Q\left(\frac{V_0'}{2\sigma_m}\right)$$

$$4) m_x = E [2 \cos(5t + \varphi)] =$$

$$= \frac{1}{2\pi} \int_{-\frac{\pi}{4}}^{\frac{7\pi}{4}} 2 \cos(5t + \varphi) d\varphi = 0$$

$$\sigma_x^2 = E [4 \cos^2(5t + \varphi)] = 4 E \left[\frac{1}{2} + \frac{1}{2} \cos(10t + 2\varphi) \right] \\ = 2$$

$$\sigma_e^2 = \frac{\Delta^2}{12} = \left(\frac{4}{L} \right)^2 \cdot \frac{1}{12}$$

$$SNR = 2 \cdot \frac{12^3 \cdot L^2}{164} = \frac{3}{2} \cdot L^2 \quad L = 128$$

5) Nella banda $[300, 3200]$ Hz si possono trasmettere

$$\frac{1}{T} = 2900 \text{ symbol/s}.$$

Per trasmettere almeno 19 kbit/s, ciascun simbolo deve trasmettere almeno 7 bit. Le costellazioni che lo fanno sono $2^7 = 128$.